Chapter Three

FERMAT'S PRINCIPLE AND ITS APPLICATIONS

Now in the further development of science, we want more than just a formula. First we have an observation, then we have numbers that we measure, then we have a law which summarizes all the numbers. But the real glory of science is that we can find a way of thinking such that the law is evident. The first way of thinking that made the law about the behavior of light evident was discovered by Fermat in about 1650, and it is called the principle of least time, or Fermat's principle.

—Richard Feynman in *Feynman Lectures on Physics*, Vol. I

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: state Fermat's principle and its modified version.
LO 2: derive laws of reflection and refraction from Fermat's principle.
LO 3: describe various phenomena by solving the ray equation.
LO 4: obtain ray paths in inhomogeneous media.
LO 5: calculate the time taken by the rays in propagating through a parabolic index medium.

Important Milestones

140 AD *Greek physicist Claudius Ptolemy measured the angle of refraction in water for different angles of incidence in air and tabulated it.*

1621 *Although the above-mentioned numerical table was made in 140 AD, it was only in 1621 that Willebrord Snell, a Dutch mathematician, discovered the law of refraction which is now known as Snell's law.*

1637 *Descartes derived Snell's law; his derivation assumed corpuscular model of light.*

1657 *Pierre de Fermat enunciated his principle of 'least time' and derived Snell's law of refraction and showed that if the velocity of light in the second medium is less, the ray would bend towards the normal, contrary to what is predicted by the 'corpuscular theory'.*

3.1 INTRODUCTION LO 1

The study of the propagation of light in the realm of geometrical optics employs the concept of rays. To understand what a ray is, consider a circular aperture in front of a point source P as shown in Fig. 3.1. When the diameter of the aperture is quite large (~ 1 cm), then on the screen SS', one can see a patch of light with well-defined boundaries. When we start decreasing the size of the aperture, then at first the size of the patch starts decreasing, but when the size of the aperture becomes very small ($\lesssim$ 0.1 mm) then the pattern obtained on SS' ceases to have well-defined boundaries. This phenomenon is known as diffraction and is a direct consequence of the finiteness of the wavelength (which is denoted by λ). In Chapters 18, 19 and 20 we will discuss the phenomenon of diffraction in great detail and will show that the diffraction effects become smaller with the decrease in wavelength and indeed in the limit of $\lambda \to 0$, the diffraction effects will be absent, and even for extremely small diameters of the aperture, we will obtain a well-defined

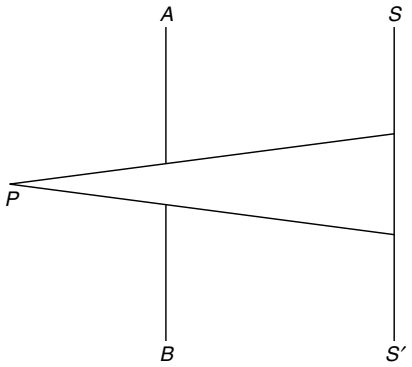


Fig. 3.1 The light emitted by the point source P is allowed to pass through a circular hole and if the diameter of the hole is very large compared to the wavelength of light, then the light patch on the screen SS' has well-defined boundaries.

shadow on the screen SS'. Therefore, in the zero wavelength limit one can obtain an infinitesimally thin pencil of light; this is called a *ray*. Thus, a ray defines the path of propagation of the energy in the limit of the wavelength going to zero. Since light has a wavelength of the order of 10^{-5} cm, which is small compared to the dimensions of normal optical instruments like lenses, mirrors, etc., one can, in many applications, neglect the finiteness of the wavelength. The field of optics under such an approximation (i.e., the neglect of the finiteness of the wavelength) is called geometrical optics.

The field of geometrical optics can be studied by using Fermat's principle which determines the path of the rays. According to this principle, *the ray will correspond to that path for which the time taken is an extremum in comparison to nearby paths*, i.e., it is either a minimum or a maximum or stationary*. Let $n(x, y, z)$ represent the position-dependent refractive index. Then

$$\frac{ds}{c/n} = \frac{n\,ds}{c}$$

will represent the time taken to traverse the geometric path ds in a medium of refractive index n. Here, c represents the speed of light in free space. Thus, if τ represents the total time taken by the ray to traverse the path AB along the curve C (see Fig. 3.2) then

$$\tau = \frac{1}{c}\sum_i n_i\ ds_i = \frac{1}{c}\int_{A\xrightarrow[C]{}B} n\,ds \qquad \textbf{(3.1)}$$

where ds_i represents the ith arc length and n_i the corresponding refractive index; the symbol $A\xrightarrow[C]{}B$ below the integral represents the fact that the integration is from

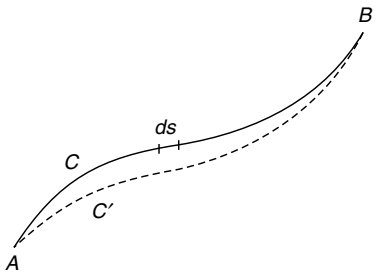


Fig. 3.2 If the path ACB represents the actual ray path then the time taken in traversing the path ACB will be an extremum in comparison to any nearby path $AC'B$.

the point A to B through the curve C. Let τ' be the time taken along the nearby path $AC'B$ (shown as the dashed curve in Fig. 3.2), and if ACB indeed represents the path of a ray, then τ will be either less than, greater than or equal to τ' for *all* nearby paths like $AC'B$. Thus according to Fermat's principle, out of the many paths connecting the two points, the light ray would follow that path for which the time taken is an extremum. Since c is a constant, one can alternatively define a ray as the path for which

$$\int_{A\xrightarrow[C]{}B} n\,ds \qquad \textbf{(3.2)}$$

is an extremum**; the above integral represents the optical path from A to B along C; i.e., the ray would follow the path for which

$$\delta \int_{A\to B} n\,ds = 0 \qquad \textbf{(3.3)}$$

where the left-hand side represents the change in the value of the integral due to an infinitesimal variation of the ray path. We may mention here that according to the original statement of Fermat:

> *The actual path between two points taken by a beam of light is the one which is traversed in the least time.*

The above statement is incomplete and slightly incorrect. The correct form is:

> *The actual ray path between two points is the one for which the optical path length is stationary with respect to variations of the path.*

This is expressed by Eq. (3.3) and in this formulation, the ray paths may be maxima, minima or stationary.

From the above principle one can immediately see that in a homogenous medium (i.e., in a medium whose refractive

* The entire field of classical optics (both geometrical and physical) can be understood from Maxwell's equations, and of course, Fermat's principle can be derived from Maxwell's equations (see Refs. 3.1 and 3.2).

** A nice discussion on the extremum principle has been given in Chapter 26 of Ref. 3.3.

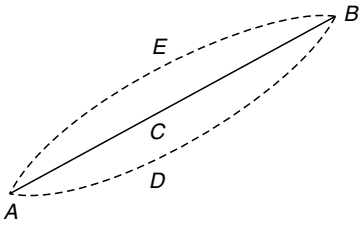


Fig. 3.3 Since the shortest distance between two points is along a straight line, light rays in a homogenous medium are straight lines; all nearby paths like AEB or ADB will take longer times.

index is constant at each point), the rays will be straight lines because a straight line will correspond to a minimum value of the optical path connecting two points in the medium. Thus referring to Fig. 3.3, if A and B are two points in a homogenous medium, then the ray path will be along the straight line ACB because any nearby path like ADB or AEB will correspond to a longer optical path.

3.2 LAWS OF REFLECTION AND REFRACTION FROM FERMAT'S PRINCIPLE LO 2

To obtain the laws of reflection and refraction from Fermat's principle, consider a plane mirror MN as shown in Fig. 3.4. We have to determine the path from A to B (via the mirror) which has the minimum optical path length. Since the path would lie completely in a homogenous medium, we need to minimize only the path length. Thus we have to find that path APB for which $AP + PB$ is a minimum. To find the position of P on the mirror, we drop a perpendicular from A on the mirror

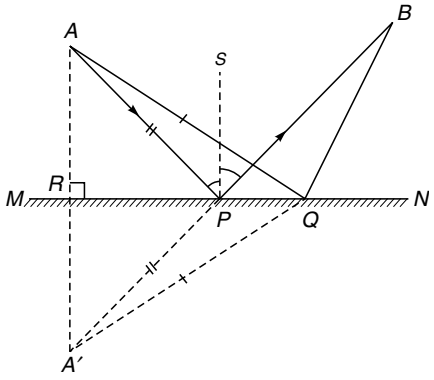


Fig. 3.4 The shortest path connecting the two points A and B via the mirror is along the path APB where the point P is such that AP, PS and PB are in the same plane and $\angle APS = \angle SPB$; PS being the normal to the plane of the mirror. The straight line path AB is *also* a ray.

and let A' be a point on the perpendicular such that $AR = RA'$; thus $AP = PA'$ and $AQ = A'Q$ where AQB is another path adjacent to APB. Thus we have to minimize the length $A'PB$. Clearly, for $A'PB$ to be a minimum, P must be on the straight line $A'B$. Thus, the points A, A', P and B will be in the same plane and if we draw a normal PS at P then this normal will also lie in the same plane. Simple geometric considerations show that

$$\angle APS = \angle SPB$$

Thus for minimum optical path length, the angle of incidence $i(= \angle APS)$ and the angle of reflection r $(= \angle SPB)$ must be equal and the incident ray, the reflected ray and the normal to the surface at the point of incidence on the mirror must be in the same plane. These form the laws of reflection. It should be pointed out that, in the presence of the mirror there will be two ray paths which will connect the points A and B; the two paths will be AB and APB. Fermat's principle tells us that whenever the optical path length is an extremum, we will have a ray, and thus, in general, there may be more than one ray path connecting two points.

To obtain the laws of refraction, let PQ be a surface separating two media of refractive indices n_1 and n_2 as shown in Fig. 3.5. Let a ray starting from the point A, intersect the interface at R and proceed to B along RB. Clearly, for minimum optical path length, the incident ray, the refracted ray and the normal to the interface must all lie in the same plane. To determine that point R for which the optical path length from A to B is a minimum, we drop perpendiculars AM and BN from A and B respectively on the interface PQ. Let $AM = h_1$, $BN = h_2$ and $MR = x$. Then since A and B are fixed, $RN = L - x$, where $MN = L$ is a fixed quantity. The optical path length from A to B, by definition, is

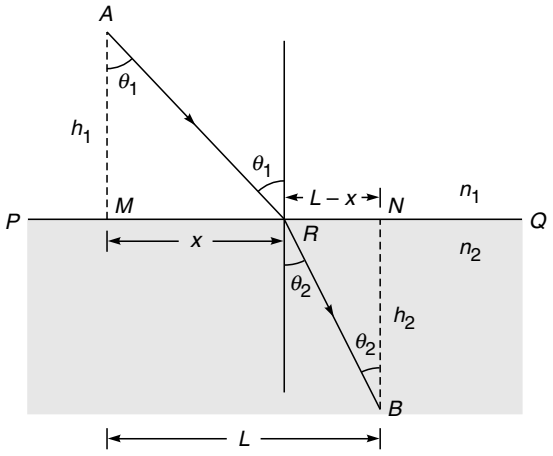


Fig. 3.5 A and B are two points in media of refractive indices n_1 and n_2. The ray path connecting A and B will be such that $n_1 \sin \theta_1 = n_2 \sin \theta_2$.

$$L_{op} = n_1\, AR + n_2\, RB$$
$$= n_1\, \sqrt{x^2 + h_1^2} \; + \; n_2 \sqrt{(L-x)^2 + h_2^2} \qquad \textbf{(3.4)}$$

To minimize this, we must have

$$\frac{dL_{op}}{dx} = 0$$

i.e., $$\frac{n_1 x}{\sqrt{x^2 + h_1^2}} - \frac{n_2 (L-x)}{\sqrt{(L-x)^2 + h_2^2}} = 0 \qquad \textbf{(3.5)}$$

Further, as can be seen from Fig. 3.5

$$\sin\, \theta_1 = \frac{x}{\sqrt{x^2 + h_1^2}}$$

and $$\sin\, \theta_2 = \frac{(L-x)}{\sqrt{(L-x)^2 + h_2^2}}$$

Thus, Eq. (3.5) becomes

$$n_1 \sin\, \theta_1 = n_1 \sin\, \theta_2 \qquad \textbf{(3.6)}$$

which is the Snell's law of refraction.

The laws of reflection and refraction form the basic laws for tracing light rays through simple optical systems, like a system of lenses and mirrors, etc.

Example 3.1 Consider a set of rays, parallel to the axis, incident on a paraboloidal reflector (see Fig. 3.6). Show by using Fermat's principle, that all the rays will pass through the focus of the paraboloid; a paraboloid is obtained by rotating a parabola about its axis. This is the reason why a paraboloidal reflector is used to focus parallel rays from a distant source, like in radio astronomy (see Figs. 3.7 and 3.8).

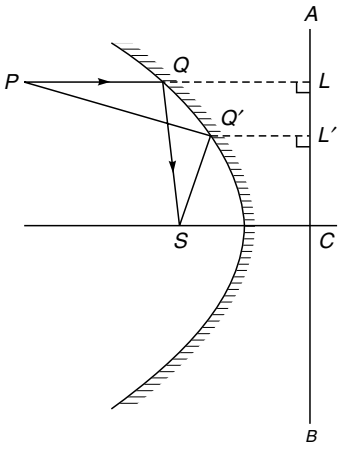


Fig. 3.6 All rays parallel to the axis of a paraboloidal reflector pass through the focus after reflection (the line *ACB* is the directrix). It is for this reason that antennas (for collecting electromagnetic waves) or solar collectors are often paraboloidal in shape.



Fig. 3.7 A paraboloidal satellite dish. [*Photograph courtesy: McGraw-Hill Digital Access Library. A color photograph of the above figure appears as Fig. 1 in the prelim pages*].



Fig. 3.8 Fully steerable 45 m paraboloidal dishes of the Giant Metrewave Radio Telescope (GMRT) in Pune, India. The GMRT consists of 30 dishes of 45 m diameter with 14 antennas in the Central Array. [*Photograph courtesy: Professor Govind Swarup, GMRT, Pune. A color photograph of the above figure appears as Fig. 2 in the prelim pages*].

Solution: Consider a ray PQ, parallel to the axis of the parabola, incident at the point Q (see Fig. 3.6). In order to find the reflected ray, one has to draw a normal at the point Q and then draw the reflected ray. It can be shown from geometrical considerations that the reflected ray QS will always pass through the focus S. However, this procedure will be quite cumbersome and as we will show, the use of Fermat's principle leads us to the desired results immediately.

To use Fermat's principle, we try to find out the ray connecting the focus S and an arbitrary point P (see Fig. 3.6). Let the ray path be $PQ'S$. According to Fermat's principle, the ray path will correspond to a minimum value of $PQ' + Q'S$. From the point Q' we drop a perpendicular $Q'L'$, on the directrix AB. From the definition of the parabola it follows that $Q'L' = Q'S$. Thus,

$$PQ' + Q'S = PQ' + Q'L'$$

Let L be the foot of the perpendicular drawn from the point P on AB. Then, for $PQ' + Q'L'$ to be a minimum, the point Q should lie on the straight line PQL, and thus the actual ray which connects the points P and S will be $PQ + QS$ where PQ is parallel to the axis. Therefore, all rays parallel to the axis will pass through S and conversely, all rays emanating from the point S will become parallel to the axis after suffering a reflection.

Example 3.2 Consider an elliptical reflector whose foci are the points S_1 and S_2 (see Fig. 3.9). Show that all rays emanating from the point S_1 will pass through the point S_2 after undergoing a reflection.

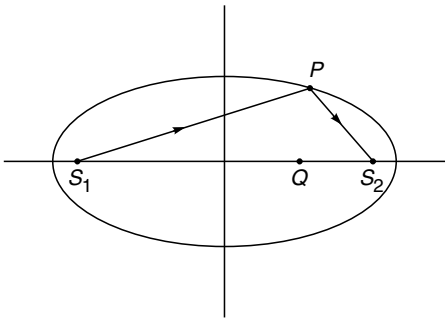


Fig. 3.9 All rays emanating from one of the foci of an ellipsoidal reflector will pass through the other focus.

Solution: Consider an arbitrary point P on the ellipse (see Fig. 3.9). It is well known that $S_1P + S_2P$ is a constant and therefore, all rays emanating from the point S_1 will pass through S_2. (Notice that here we have an example where the time taken by the ray is stationary, i.e., it is neither a maximum nor a minimum but has a constant value for all points lying on the mirror.) As a corollary, we may note the following two points:

(a) Excepting the rays along the axis, no other ray (emanating from either of the foci) will pass through an arbitrary point Q which lies on the axis.
(b) The above considerations will remain valid even for an ellipsoid of revolution obtained by rotating the ellipse about its major axis.

Because of the above-mentioned property of elliptical reflectors, they are often used in laser systems. For example, in a ruby laser (see Sec. 27.3) one may have a configuration in which the laser rod and the flash lamp coincide with the focal lines of a cylindrical reflector of elliptical cross section; such a configuration leads to an efficient transfer of energy from the lamp to the ruby rod.

Example 3.3 Consider a spherical refracting surface SPM separating two media of refractive indices n_1 and n_2 (see Fig. 3.10). The point C represents the center of the spherical surface SPM. Consider two points O and Q such that the points O, C and Q are

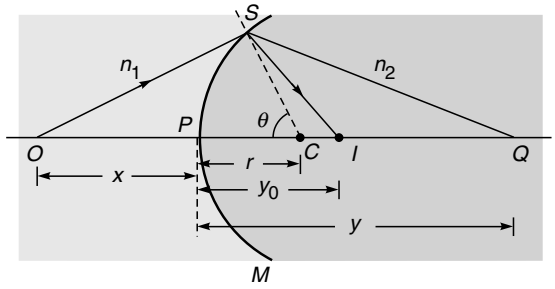


Fig. 3.10 SPM is a spherical refracting surface separating two media of refractive indices n_1 and n_2. C represents the center of the spherical surface.

in a straight line. Calculate the optical path length OSQ in terms of the distances x, y, r and the angle θ (see Fig. 3.10). Use Fermat's principle to find the ray connecting the two points O and Q. Also, assuming the angle θ to be small, determine the paraxial image of the point O.

[*Note*: We reserve the symbol R to represent the radius of curvature of a spherical surface which will be positive (or negative) depending upon whether the center of curvature lies on the right (or left) of the point P. The quantity r represents the magnitude of the radius of curvature which, for Fig. 3.10, happens to be R. Similarly, the quantities x and y are the magnitudes of the distances; the sign convention is discussed later on in this problem.]

Solution: From the triangle SOC, we have

$$OS = [(x + r)^2 + r^2 - 2(x + r)\, r \cos\theta]^{1/2}$$

$$\approx \left[x^2 + 2rx + 2r^2 - 2(xr + r^2)\left(1 - \frac{\theta^2}{2}\right)\right]^{1/2}$$

$$\approx x\left[1 + \frac{rx + r^2}{x^2}\theta^2\right]^{1/2} \approx x + \frac{1}{2}r^2\left(\frac{1}{r} + \frac{1}{x}\right)\theta^2$$

where we have assumed θ (measured in radians) to be small so that we may use the expression

$$\cos\theta \approx 1 - \frac{\theta^2}{2}$$

and also make a binomial expansion. Similarly, by considering the triangle SCQ, we would have

$$SQ \approx y - \frac{1}{2}r^2\left(\frac{1}{r} - \frac{1}{y}\right)\theta^2$$

Thus the optical path length OSQ is given by

$$L_{op} = n_1\, OS + n_2\, SQ$$

$$\approx (n_1 x + n_2 y) + \frac{1}{2}r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta^2 \qquad (3.7)$$

For the optical path to be an extremum we must have

$$\frac{dL_{op}}{d\theta} = 0 = r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]\theta \qquad (3.8)$$

Thus, unless the quantity inside the square brackets is zero we must have $\theta = 0$ implying that the *only* ray connecting the points O and Q will be the straight line path OPQ which also follows from Snell's law because the ray OP hits the spherical surface normally and should proceed undeviated.

On the other hand, if the value of y was such that the quantity inside the square brackets was zero, i.e., if y was equal to y_0 such that

$$\frac{n_2}{y_0} + \frac{n_1}{x} = \frac{n_2 - n_1}{r} \qquad \textbf{(3.9)}$$

then $dL_{op}/d\theta$ would vanish for *all* values of θ; of course, θ is assumed to be small—which is the paraxial approximation. Now, if the point I corresponds to $PI = y_0$ (see Fig. 3.10) then *all* paths like OSI are allowed ray paths implying that *all* (paraxial) rays emanating from O will pass through I and I will, therefore represent the paraxial image point. Obviously, all rays like OSI (which start from O and pass through I) take the *same* amount of time in reaching the point I.

We should mention that Eq. (3.9) is a particular form of the equation determining the paraxial image point

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \qquad \textbf{(3.10)}$$

with the sign convention that all distances measured to the right of the point P are positive and those to its left negative. Thus $u = -x$, $v = +y$ and $r = +R$.

In order to determine whether the ray path OPQ corresponds to minimum time or maximum time or stationary, we must determine the sign of $d^2L_{op}/d\theta^2$ which is given by

$$\frac{d^2L_{op}}{d\theta^2} = r^2\left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r}\right]$$

$$= r^2 n_2\left[\frac{1}{y} - \frac{1}{y_0}\right]$$

Obviously, if $y > y_0$ (i.e., the point Q is on the right of the paraxial image point I) $d^2L_{op}/d\theta^2$ is negative and the ray path OPQ corresponds to *maximum* time in comparison with nearby paths and conversely. On the other hand, if $y = y_0$, $d^2L_{op}/d\theta^2$ will vanish implying that the extremum corresponds to stationarity. Thus, in the paraxial approximation, all rays emanating from the point O will take the same amount of time in reaching the point I.

Alternatively, one can argue that if I is the paraxial image point of P then

$$n_1\, OP + n_2\, PI = n_1\, OS + n_2\, SI$$

Thus, when Q lies on the right of the point I, we have

$$n_1\, OP + n_2\, PQ = n_1\, OS + n_2\,(SI - PI + PQ)$$
$$= n_1\, OS + n_2\,(SI + IQ)$$
$$> n_1\, OS + n_2\, SQ$$

implying that the ray path OPQ corresponds to a maximum. Similarly, when Q lies on the left of the point I then the ray path OPQ corresponds to a minimum and when Q coincides with I, we have the stationarity condition.

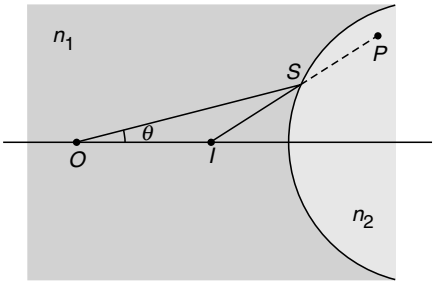


Fig. 3.11 The refracted ray is assumed to diverge away from the principal axis.

Example 3.4 We again consider refraction at a spherical surface; however, the refracted ray is assumed to diverge away from the principal axis (see Fig. 3.11). Let us consider paraxial rays and let I be a point (on the axis) such that $n_1\, OS - n_2\, SI$ is independent of the point S. Thus, for paraxial rays, the quantity

$$n_1\, OS - n_2\, SI \qquad \text{is independent of } \theta \qquad \textbf{(3.11)}$$

and is an extremum. Let P be an arbitrary point in the second medium and we wish to find the ray path connecting the points O and P. For OSP to be an allowed ray path

$$L_{op} = n_1\, OS + n_2\, SP \text{ should be an extremum}$$

or

$$L_{op} = (n_1\, OS - n_2\, SI) + n_2\,(IS + SP) \text{ should be an extremum}$$

where we have added and subtracted $n_2\, SI$. Now, the point I is such that the first quantity is already an extremum thus, the quantity $SP + SI$ should be an extremum and therefore it should be a straight line. Thus the refracted ray must appear to come from the point I. We may therefore say that for a virtual image, we must make the quantity

$$n_1\, OS - n_2\, SI \qquad \textbf{(3.12)}$$

an extremum.

3.3 RAY PATHS IN AN INHOMOGENEOUS MEDIUM

LO 3

In an inhomogeneous medium, the refractive index varies in a continuous manner and, in general, the ray paths are curved. For example, on a hot day, the air near the ground has a higher temperature than the air which is much above the surface. Since the density of air decreases with increase in temperature, the refractive index increases continuously as we go above the ground. This leads to the phenomenon known as *mirage*. We will use Snell's law (or Fermat's principle) to determine the ray paths in an inhomogeneous medium. We will restrict ourselves to the special case when the refractive index changes continuously along one direction only; we assume this direction to be along the x-axis.